Energy Asset Risk Assessment

& Predictive Modelling Dashboard Report

Client-facing analytical report and dashboard walkthrough

Prepared for stakeholder review

Project package: Machine Learning Model, Technical Whitepaper, Power BI Dashboard, and Analysis Report

Sector	Energy / asset risk analytics
Primary objective	Prioritise assets by relative operational and financial risk
Core outputs	Risk score, risk level, claim probability, and expected loss
Recommended model	Gradient Boosting, based on stronger overall performance

1. Executive Summary

This report summarises the final Power BI dashboard and the analytical conclusions derived from the underlying simulation-based machine learning workflow. The solution was built to answer three stakeholder questions: where risk and expected loss are concentrated, how reliable the model is, and which factors are driving asset risk.

Across the portfolio, expected loss is concentrated in a relatively small number of fossil-fuel assets and countries. Coal and gas assets dominate expected financial exposure, while the United States and China represent the largest country-level concentrations. The dashboard also shows that most assets are classified as low risk, with a much smaller subset of medium- and high-risk assets requiring greater attention.

From a modelling perspective, Gradient Boosting is selected as the primary model for risk scoring, while Random Forest performs slightly better for classification tasks. The model should therefore be used as a prioritisation and screening tool rather than as a deterministic predictor of specific asset failures.

- Most assets fall into the low-risk category, but a smaller high-priority subset drives disproportionate expected loss.
- Expected loss is concentrated in coal and gas assets, with meaningful geographic concentration in a limited number of countries.
- The regression model performs strongly (R^2 = 0.86; RMSE = 4.34), supporting credible risk ranking across the portfolio.
- The classification model is moderate rather than perfect (ROC AUC = 0.65; Recall = 0.56), making it suitable for screening rather than exact event prediction.
- After excluding the engineered fuel weighting variable from the driver view, asset age and capacity interactions emerge as the most important operational drivers.

2. Dashboard Structure and Interpretation

The dashboard is organised into three pages, each serving a distinct stakeholder purpose.

Page	Primary question answered	Stakeholder value
1. Energy Asset Risk Overview	Where are risk and expected loss concentrated?	Immediate portfolio view of scale, distribution, concentration, and asset clustering.
2. Model Performance	How strong is the modelling approach?	Evidence that the selected model is fit for ranking, screening, and supporting decisions.
3. Risk Drivers & Key Factors	Why are some assets riskier than others?	Explains model behaviour and links risk back to interpretable asset characteristics.

3. Page 1 — Energy Asset Risk Overview

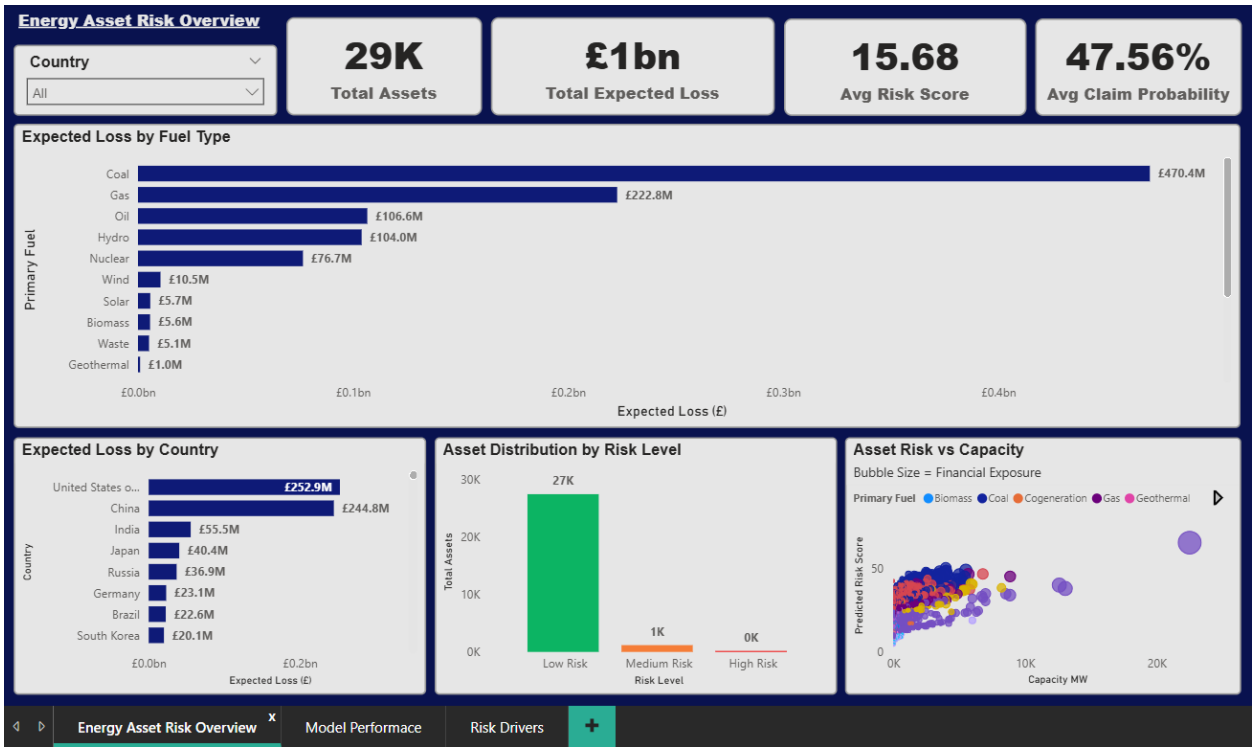

Figure 1. Executive dashboard page: Energy Asset Risk Overview.

The first page is the portfolio-level summary. It is designed for senior stakeholders who need a fast understanding of exposure, concentration, and relative risk without needing to interpret the underlying model mechanics.

3.1 KPI cards

KPI	Value shown	Plain-English meaning
Total Assets	29K	The portfolio contains approximately 29,000 assets. This establishes the scale of the analysis and confirms that the dashboard is evaluating a broad, multi-asset portfolio rather than a narrow sample.
Total Expected Loss	£1bn	This is the aggregate modelled financial exposure across the full portfolio. It does not mean £1bn will definitely be lost; rather, it estimates the amount of exposure that should be prioritised for monitoring and mitigation.
Avg Risk Score	15.68	This is the average modelled asset risk score on the 0–100 scale. In simple terms, the average asset sits in the lower-risk part of the portfolio, which is consistent with the risk-band chart below.

Avg Claim Probability	47.56%	This is the average modelled event probability across the portfolio. It should be interpreted as a relative screening signal, not a real insurance probability, because the underlying targets are synthetic.

3.2 Chart-by-chart interpretation

Expected Loss by Fuel Type. This chart shows where aggregate financial exposure is concentrated by fuel category. Coal is the dominant exposure at roughly £470M, followed by gas at about £223M. This immediately indicates that the largest financial concentration in the portfolio is associated with fossil-fuel assets.

Expected Loss by Country. This chart highlights geographic concentration of modelled loss. The United States and China are materially ahead of all other countries, indicating that exposure is not evenly distributed geographically. For portfolio management, these countries would represent priority areas for more detailed review.

Asset Distribution by Risk Level. The portfolio is overwhelmingly low risk, with only a much smaller number of medium- and high-risk assets. This is a healthy pattern for a broad asset portfolio but also reinforces the need to identify the smaller subset that drives disproportionate risk and cost.

Asset Risk vs Capacity. The scatter plot provides an asset-level view of the relationship between asset size, risk score, and financial exposure. Larger bubbles indicate greater expected loss. The visual shows that higher-capacity assets can create outsized financial exposure even when their risk scores are only moderate, which is an important management insight.

3.3 Stakeholder insight from Page 1

- Risk and cost are concentrated rather than evenly distributed across the portfolio.
- Fuel type and geography are major portfolio segmentation lenses for decision-making.
- A relatively small number of larger assets deserve disproportionate attention due to their financial impact.
- This page is best used for prioritisation, portfolio review, and executive discussion.

4. Page 2 — Model Performance

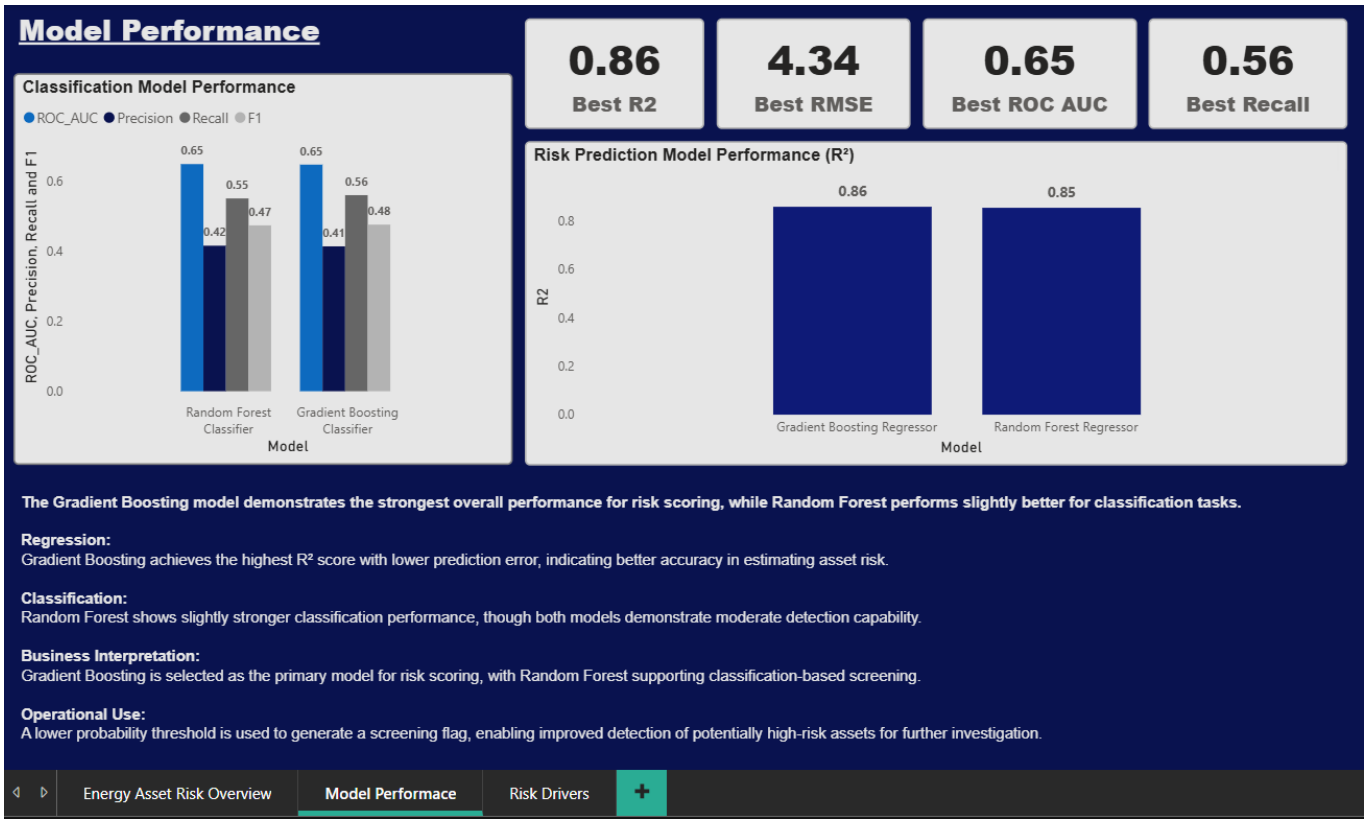

Figure 2. Model Performance page.

The second page is the technical assurance page. Its job is to make the dashboard credible by showing how well the models perform and which model should be trusted for the final business-facing outputs.

4.1 KPI cards

KPI	Value shown	Interpretation
Best R^2	0.86	The selected regression model explains approximately 86% of the variance in risk scores. In simple terms, it is strong at estimating relative asset risk and supports ranking assets by risk level.
Best RMSE	4.34	This is the average size of the regression error on a 0–100 risk scale. An error of around four points is relatively small, which supports confidence in the risk score outputs.
Best ROC AUC	0.65	This measures the classification model's ability to distinguish higher-risk from lower-risk cases. A value of 0.65 is better than random and suitable for screening, but it is not strong enough to be described

		as precise event prediction.
Best Recall	0.56	The model captures around 56% of positive cases. In practical terms, it identifies more than half of the assets flagged as risky in the synthetic target, but some will still be missed.

4.2 Chart-by-chart interpretation

Classification Model Performance. This chart compares Random Forest and Gradient Boosting across ROC AUC, Precision, Recall, and F1. The main message is that both models are workable, but Gradient Boosting offers the stronger overall balance. It performs at least as well on ROC AUC and better supports the chosen business objective of ranking and screening.

Risk Prediction Model Performance (R^2). This chart compares the regression models directly. Both models perform well, but Gradient Boosting is marginally stronger. The difference is not huge, which is a good sign. It implies the signal is stable rather than dependent on one highly unstable algorithm.

Model Selection Insight. The text box translates technical metrics into business meaning. It explains why Gradient Boosting was selected and clarifies that the model should be used to prioritise and screen assets rather than to predict exact outcomes with certainty.

4.3 Stakeholder insight from Page 2

- The risk scoring model is strong enough to support prioritisation decisions.
- The event-detection model is moderate and should be used for screening rather than definitive prediction.
- Gradient Boosting is the best overall model because it offers the strongest balance of performance and interpretability.

5. Page 3 — Risk Drivers & Key Factors

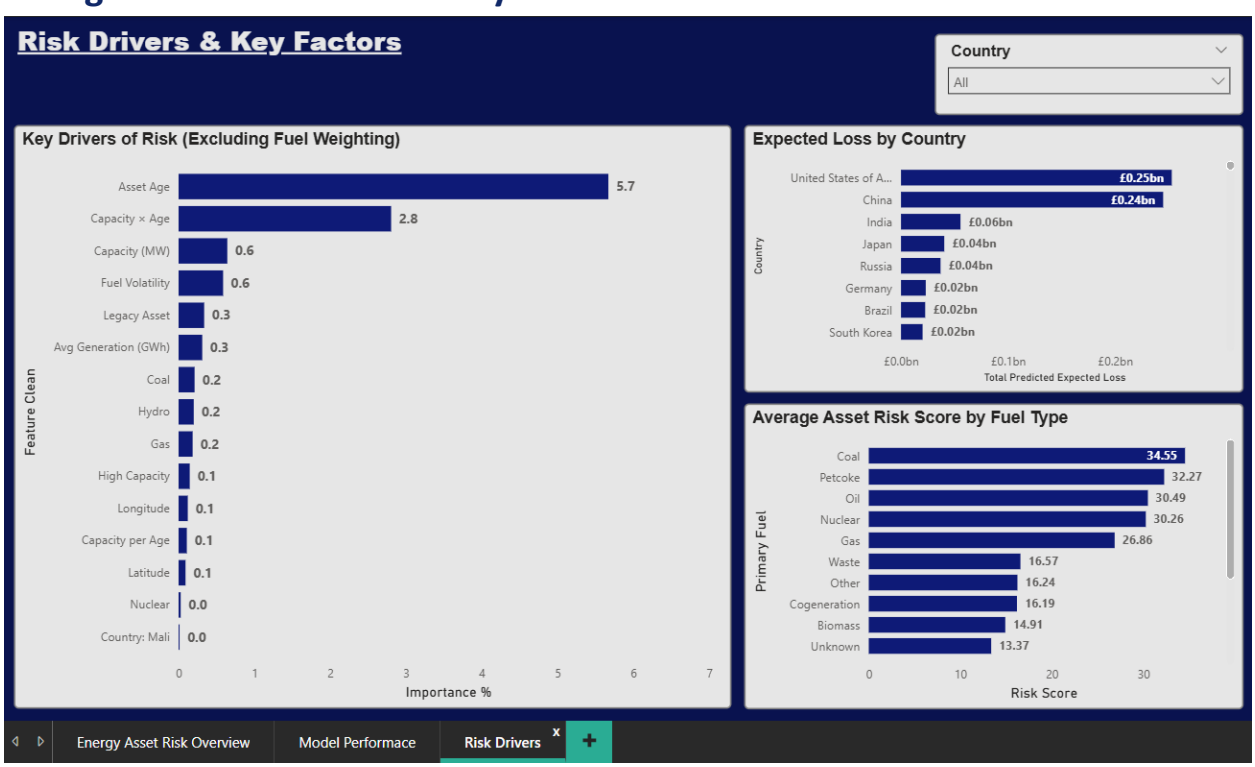

Figure 3. Risk Drivers & Key Factors page.

The third page explains why the dashboard is assigning higher risk to some assets than others. This is especially useful for stakeholders who need a defensible narrative rather than a black-box score.

5.1 Main driver chart

Key Drivers of Risk (Excluding Fuel Weighting). This chart shows the top 15 regression model features after intentionally removing the engineered fuel weighting variable. That exclusion is analytically sensible because fuel weighting was deliberately built into the synthetic target and would otherwise dominate the view. By removing it, the chart highlights the next-most-important operational drivers.

The most important driver is Asset Age, followed by Capacity × Age. This means that older assets, especially when combined with scale, are the most influential operational contributors to risk. Capacity, fuel volatility, and legacy asset status also matter, but to a much smaller degree.

This is a strong business narrative: the model is not simply rewarding or penalising one category at random. It is responding to a coherent combination of asset maturity, size, and structural stress factors.

5.2 Supporting charts

Expected Loss by Country. This supporting chart reinforces the concentration view from Page 1, but it is placed alongside the driver chart to help stakeholders connect model logic with financial exposure. It shows where the business impact sits geographically.

Average Asset Risk Score by Fuel Type. This chart explains which fuel categories tend to sit higher on the risk spectrum. Coal, petcoke, oil, and nuclear appear materially higher than lower-risk categories such as biomass or unknown. This helps translate the model into operational intuition: fuel mix remains an important lens for understanding risk.

5.3 Stakeholder insight from Page 3

- The model is primarily driven by asset age and the interaction between age and capacity.
- Fuel-related effects remain visible even after excluding the engineered fuel weighting variable.
- The dashboard provides a clear explanation layer, which strengthens trust in the model outputs.

6. Overall Analytical Conclusions

Taken together, the three dashboard pages tell a coherent story.

- Page 1 identifies where financial exposure and risk are concentrated.
- Page 2 demonstrates that the selected model is robust enough to support prioritisation and screening.
- Page 3 explains why the model assigns higher risk to certain assets.

This sequence is valuable because it mirrors how stakeholders make decisions: first identify the problem, then establish confidence in the evidence, then understand the underlying drivers.

7. Recommendations

- Use the dashboard as a prioritisation tool for asset review, maintenance planning, and exposure monitoring.
- Focus immediate investigation on older, larger, and higher-volatility assets, especially in high-exposure fuel categories and countries.
- Treat event probability and expected loss as decision-support metrics rather than precise forecasts.
- If this work were moved toward production, the next step would be to replace synthetic targets with real incident, outage, claims, or maintenance history data.

8. Limitations and Appropriate Use

This report should be read with one important caveat: the modelling workflow is simulation-based. The targets used to train the models were engineered to create a realistic portfolio analytics exercise, but they are not real-world claims or failure outcomes.

- Risk Score is a relative prioritisation metric, not a guaranteed event forecast.
- Claim Probability is a modelled screening signal, not an actuarial insurance probability.
- Expected Loss is a modelled exposure estimate, not an audited accounting figure.

These limitations do not reduce the value of the project as a portfolio-quality analytical solution. They simply define the correct level of interpretation and keep the work professionally honest.

9. Conclusion

This project delivers a comprehensive, end-to-end analytical solution that combines machine learning with intuitive business intelligence. The integration of predictive modelling and a well-structured Power BI dashboard enables stakeholders to move beyond raw data and towards informed, actionable decision-making.

The strength of the solution lies not only in the technical implementation, but in its ability to translate complex model outputs into clear, practical insights. Risk is not only quantified, but contextualised — allowing decision-makers to understand where exposure exists, what is driving it, and how best to respond.

Overall, the analysis provides a robust foundation for prioritising high-risk assets, improving resource allocation, and supporting a more proactive approach to risk management across the energy asset portfolio.